In-Car Vital Sign Detection and Respiratory Rate Estimation Using NLS-Modified I/Q Radar Signals

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***Abstract*—This paper presents a machine learning-based binary classification approach for detecting the presence of living occupants in a car using radar signals. Unlike traditional noise-sensitive peak detection techniques or deep learning methods, which require significant computational resources, our solution utilizes a lightweight and efficient machine learning model. Additionally, to enhance signal quality prior to classification and then breathing rate estimation, we introduce a mathematically closed-form DC offset removal method based on a nonlinear least-squares (NLS) approach, which geometrically corrects the radar's I/Q signal to better reveal subtle chest displacements causing the breathing rate. The proposed system is implemented using Infineon's BGT60TR13C radar and experimentally validated in a car environment. The results demonstrate the effectiveness of this novel approach for in-car life detection, and breathing rate estimation, which supports critical applications such as post-crash rescue and prevention of child heatstroke incidents.**

***Index Terms*—mmWave radar, FMCW, vital sign monitoring, in-car sensing, life detection, DC offset removal, nonlinear least squares, I/Q signal processing, machine learning, automotive safety.**

I. INTRODUCTION

Due to the versatility of radars and their ability to measure contactlessly, they can be used for in-car applications such as gesture recognition [1], drowsiness detection [2], vital sign monitoring [3]–[7], occupancy detection [8], and child detection [9]–[12]. In these studies, various types of radar have been used, such as the Ultra Wideband (UWB) radar, Doppler radar, and Frequency Modulated Continuous Wave (FMCW) radar. However, considering the ability to select the efficient range bin, low sampling rate, and low power consumption, FMCW is a reasonable choice for the in-car application.

In [10], passenger monitoring is performed using classification of the type of occupant. The classes considered are an empty car, a car with an adult on board, or a car with a child on board. The method used in the study mentioned is convolutional Long Short-term Memory (LSTM). However, some objects with gradual nuance movement can produce a pattern that can pose a challenge of false positives in the classification, which is worth investigating. The binary classification of alive occupancy in the car not only can help prevent forgetting an animal or child within the car [13], [14], but also can be helpful in post-crash analysis to send data to a medical center. Then, after the detection of a live presence, monitoring of vital signs can occur.

In this research, our novel contribution is to use a machine learning algorithm with a small number of datasets for a binary classification task to detect the life presence within the car. In this context, one of the key challenges is the presence of DC offsets, which are introduced by imperfections in the radar system, sensor drift, or environmental factors. DC offsets shift the signal in the In-phase (I) and Quadrature (Q) planes, making it difficult to accurately analyze the radar echo to detect and estimate vital signs, such as breathing rate. Thus, we propose a method for DC offset removal based on a Nonlinear Least Squares (NLS) approach [1], which minimizes the geometric deviation of the signal from a perfect circle in the I/Q plane. This approach ensures that the radar's slow-time signal is modified and prepared for life detection among alive and non-living classes and improves the accuracy of the estimation. Moreover, the DC removal approach mentioned above is independent of radar configuration in contrast to the high-pass filter, which was investigated in our previous work [15].

To validate our results, we utilize the BGT60TR13C [16] from Infineon, which operates at 60 GHz frequency with 5.5 GHz instantaneous bandwidth as a Multiple-Input Single-Output (MISO) FMCW radar. The system is installed in a car, between the front seats, to monitor the occupant's chest displacement as part of an in-car life classification system on the back seat. The periodic motion of the passenger's chest during breathing is detected using the proposed radar signal processing pipeline.

The remainder of the paper is organized as follows. In Section II, we describe the proposed signal processing pipeline used to extract and analyze the radar data, remove DC[2], and then perform the classification. Experimental results are provided in Section III, followed by conclusions in Section IV.

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[1]In this study, this method is not used as a spectrum analysis approach for breathing rate estimation.

[2]**Notations** Throughout this paper, scalars are denoted by italic lowercase letters (e.g., x), vectors by bold lowercase letters (e.g., $\mathbf{x}$), and matrices by bold uppercase letters (e.g., $\mathbf{X}$). Sets of real and complex numbers are represented by $\mathbb{R}$ and $\mathbb{C}$, respectively. Also, $\odot$ denotes element-wise multiplication.

II. Proposed Method

In this research, we used the BGT60TR13C [16] from Infineon Company, which operates as an FMCW MISO radar. The FMCW radar is installed on the inner roof between the front seats to monitor the position of the middle back seat and detect life occupancy. The displacement of the passenger's chest wall, which produces periodic movement, is modeled as $\phi_d(t) = \frac{4\pi(d_0+x(t))}{\lambda}$. Here, d_0 is the initial distance between the subject and the radar at time $t = 0$, and $x(t)$ represents the movement of the surface of the object (such as the expansion of the chest during breathing) as a function of time. λ is the wavelength of the radar, given by $\lambda = \frac{c}{f}$, where c is the speed of light, and f is the operating frequency of the radar.

The schematic of the system model and the configuration of the radar within the car, with an initial distance of d_0 from the target, is shown in Fig. 1. After the radar signal passes through the de-chirp circuit and is sampled at 1 MHz by the analog to digital converter (ADC), a range-Fast Fourier Transform (FFT) is performed, followed by an angle-FFT across the three receive (Rx) antennas of the BGT60TR13C, forming the radar's receive beam in the direction of the passenger. The process is then followed by range-bin selection, which identifies the range bin corresponding to the object under test. The signal corresponding to this selected range bin is extracted over multiple radar returns. It is referred to as the slow-time complex I and Q signal, represented by $I[n]$ and $Q[n]$, where $n = 1, 2, \ldots, N$ and N is the number of reflected pulses. This slow-time signal, which represents the radar echo over time, is then extracted for further processing [15].

The extracted signal often contains a DC offset caused by imperfections in the radar system, sensor drift, or environmental conditions that should be processed. The offsets in the I and Q signals shift the ideal location of the signal in the I/Q plane, complicating subsequent signal analysis, especially for detecting small physiological movements such as chest displacement due to breathing. To address this issue, the DC offset needs to be removed to ensure accurate life detection and breathing rate estimation in case that any life presence is detected.

A. DC Offset Removal Using Nonlinear Least Squares

After classification and detecting alive targets, the DC offset removal is achieved using a NLS optimization approach. We assume that the signal ideally lies on a circle in the I/Q plane. However, because of DC offsets (d_I, d_Q), the center of the circle is deviated from the origin. The goal is to estimate the shifted center (d_I, d_Q) of the circle and to eliminate these offsets.

Assuming the samples lie on a circle centered at (d_I, d_Q), we have $(I[n]-d_I)^2+(Q[n]-d_Q)^2 = r^2$, for all n, where r is the (unknown) radius of the circle. This means that in order to minimize the DC offset, one can solve to the following NLS optimization problem:

$$\min_{d_I, d_Q, r} \sum_{n=1}^{N} \left((I[n] - d_I)^2 + (Q[n] - d_Q)^2 - r^2 \right)^2, \quad (1)$$

which is a nonconvex, multivariable optimization problem. Linearization is a promising solution for it. Expanding the ideal circular equation:

$$I[n]^2 + Q[n]^2 = 2d_I I[n] + 2d_Q Q[n] + c, \quad (2)$$

where $c = r^2 - d_I^2 - d_Q^2$ is an additional scalar parameter. Define the data vector: $\mathbf{b} = \mathbf{I} \odot \mathbf{I} + \mathbf{Q} \odot \mathbf{Q}$, $\mathbf{A} = \begin{bmatrix} 2\mathbf{I} & 2\mathbf{Q} & \mathbf{1} \end{bmatrix}$, where $\mathbf{1} \in \mathbb{R}^N$ is an all-ones vector and $\mathbf{I} = [I[1], \ldots, I[N]]^T$ and $\mathbf{Q} = [Q[1], \ldots, Q[N]]^T$. The least-squares solution is obtained from $\mathbf{Ax} = \mathbf{b}$, $\mathbf{x} = [d_I \ d_Q \ c]^T$. The problem in here would turn to the standard least squares, where its solution is given by

$$\hat{\mathbf{x}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}, \quad (3)$$

from which we extract the estimated DC offsets: $\hat{d}_I = \hat{x}_1$, $\hat{d}_Q = \hat{x}_2$. The DC-corrected signal components are then obtained as: $I_c[n] = I[n] - \hat{d}_I$, $Q_c[n] = Q[n] - \hat{d}_Q$, and the corrected complex signal is $z_c[n] = I_c[n] + jQ_c[n]$. This proposed approach is a closed-form solution for the circle fit, which is efficient and suitable when the problem is linearizable.

B. Life Occupancy Detection

After DC removal stage, to classify the presence of a living occupant, we use the amplitude of the complex radar signal $|z_c[n]|$ as the primary signal for feature extraction. This signal captures small periodic variations corresponding to chest movement caused by breathing, which is typically present in living subjects but absent in non-living ones.

Due to the periodic behavior of chest wall movement, from the slow-time signal, we extract variance as a statistical feature since it represents the fluctuation intensity of the signal over time.

For classification, we use a common machine learning method: **k-Nearest Neighbor (kNN)**, which is a non-parametric method that classifies samples based on the majority of the labels among their k closest neighbors in the feature space. To ensure generalization, given the limited dataset size, a 5-fold cross-validation strategy is used for model training and parameter selection. In particular, the optimal value of k in kNN is determined via validation error minimization.

III. Results

This study evaluates several in-car scenarios to distinguish between alive and non-living entities using radar sensing to later estimate the breathing rate of the life subjects. The alive class includes pseudo-breathing of a 10-month-old infant, simulated using a breathing pump installed in the torso of a doll, as well as human participants seated in a stationary car. In contrast, the non-living class consists of a half-full 5 liter bottle of water in a hand-shaken car, a compressed thermal blanket, and an empty seat. The radar sensor is mounted centrally in the car, approximately between the two front seats, oriented at a 32° angle relative to the car's roof, as illustrated in Fig. 1. The subject under observation is positioned in the middle of

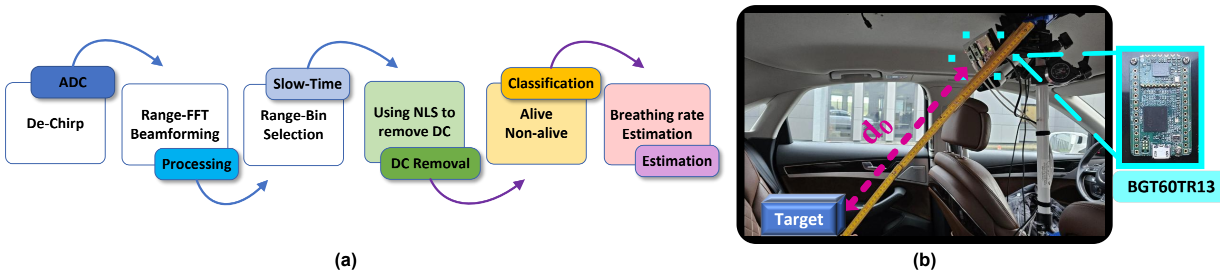


Fig. 1: (a) Block diagram of the proposed signal processing pipeline for life classification. The system includes de-chirping, ADC sampling, range and angle FFTs, slow-time bin selection, and NLS DC removal for classification followed by breathing estimation, (b) the radar in the car with an initial distance of d_0 from the target.

the back seat to ensure a consistent and unobstructed field of view.

A. DC Offset Removal

To illustrate the impact of DC offset removal for breathing rate estimation in life class, we consider a representative dataset labeled `10month_child_Engine_On_AC_Off`, which corresponds to measurements taken from the pseudo-breathing pump simulating an infant's respiration, while the car engine was running and the Air Conditioner (AC) system was turned off. Fig. 2 displays the constellation of the slow-time radar signal in two stages, including prior to DC removal, and after applying three different preprocessing methods—mean subtraction, high-pass filtering, and the proposed NLS approach, and also the phase signal of these cases. The raw signal exhibits an offset to the $(0, 0)$ center and spread, while the signal processed via NLS is better aligned to the origin and shows improved compactness. This improvement reflects more consistent phase stability and trend alleviation in the phase, which is critical for downstream micro-motion analysis. Although one example signal is shown here, the enhancement is observed across the entire dataset.

A similar improvement is observed in the unwrapped phase signal, as shown in Fig. 3. In (a), the raw data phase exhibits low-frequency fluctuations over time, which can introduce distortions and spurious trends in subsequent signal processing stages. The unwanted trend is attenuated using methods such as mean subtraction, high-pass filtering, and the NLS method. Fig. 3 (b), shows that the proposed NLS method and mean subtraction reduce the trend while not omitting the high frequency information of the signal as highpass filter does, so in the frequency domain, the detected peak representing the frequency of the breathing rate using these two methods are more accurate and close to the ground truth value captured by Neulog belt [17] shown in this plot.

B. Life Occupancy Detection

To evaluate the effectiveness of the proposed life occupancy classification approach under realistic in-car conditions, we consider a range of representative scenarios. These include: (i) a stationary car with the engine turned off, (ii) a stationary car with the engine on and AC off, and (iii) a static car

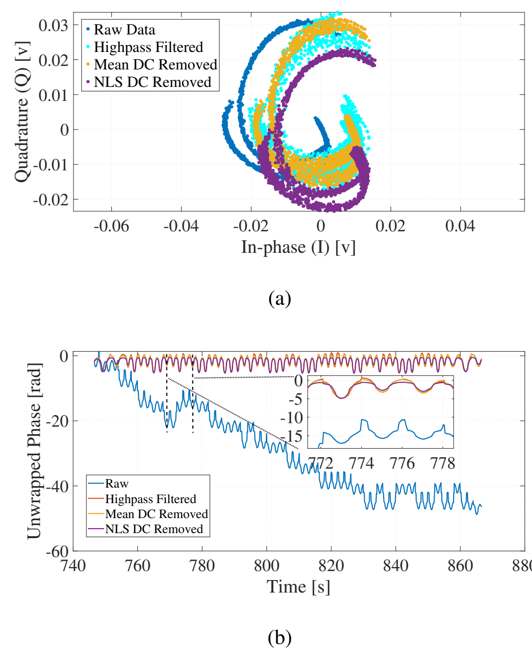
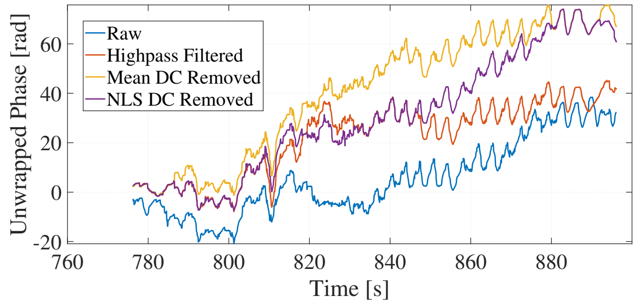


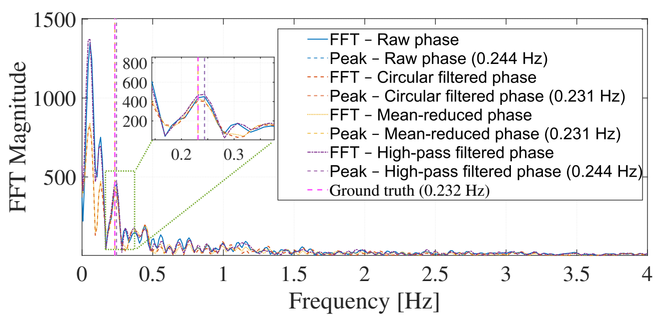
Fig. 2: (a) Constellation, and (b) phase of the complex raw signal, and the preprocessed one.

with both the engine and AC turned on. The total number of signals as data in each class of alive and non-living is 400, and the length of each signal sampled with 20 Hz for 20 s within each set is also 400. To distinguish between alive and non-living occupants seated in the rear middle position, the signal magnitude $|I_c[n] + jQ_c[n]|$ is employed as the basis for feature extraction. This magnitude signal captures the dynamic variations in the radar return purified by the previous DC removal stage, and has a clear periodic pattern in the presence of a living subject, corresponding to physiological motions such as breathing. Such a pattern is notably absent or highly attenuated in non-living scenarios, forming the key discriminative element for classification.

Fig. 4 presents representative examples of the histogram distributions of the DC-removed slow-time signal magnitudes for both alive and non-living classes. As a result of comparing the classes above, it is concluded that the breathing pattern of

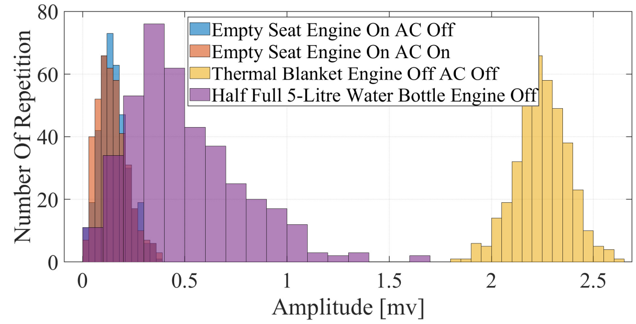


(a)

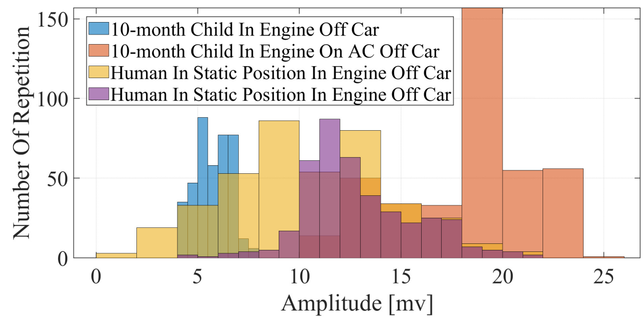


(b)

Fig. 3: The comparison of the raw signal and preprocessed signal in (a) time domain, and (b) frequency domain (for a 20 s portion of time).



(a)



(b)

Fig. 4: Histogram of samples of the $|I_c[n] + jQ_c[n]|$ signal for the classes of (a) non-living measurements, (b) alive measurements.

the alive class is different from that of non-living objects due to their statistical characteristics, as is visible in their histograms. It is worth mentioning the notion [18] that the histograms of the amplitude of the non-living class resemble a Rayleigh distribution due to its essence of noisy characteristics, mainly in cases like empty seats, as we do not have any target, and the only present phenomenon is the environmental noise. However, for living classes, apart from environmental noise, they contain periodic breathing movements, resulting in a greater amount of variance in the amplitude signal. So, due to the high variation of the amplitude of each example in the alive class and the low amount of amplitude variation in the non-living class, the variance of each time series is chosen as features for classifying alive and non-living targets.

Using kNN [19] [20], human breathing can be detected, and the result of this classifier is indicated using its confusion matrix in Table II and the learning curve of the method in Table I. Moreover, based on the figure mentioned, the optimal number of neighborhoods in the kNN method in this problem using the K-fold validation error with 5 folds [21] is 3. K-fold validation method is used when the number of data points is low; however, it should be investigated how well the model will perform when encountering new data. As a result, the kNN method is more accurate and able to generalize the model based on this data set according to its learning curve.

TABLE I: Learning behavior of kNN classifier

Train Size	Train Acc. (%)	Val. Acc. (%)
10	90	50
220	99.55	98.13
430	99.30	98.13
640	99.06	**98.75**

TABLE II: kNN confusion matrix

kNN Confusion Matrix	True Negative	True Positive
Predicted Negative	100%	0%
Predicted Positive	2.5%	97.5%

IV. Conclusion

In this study for further enhancement of the accuracy of the life detection and estimation of breathing rate, a DC offset removal algorithm based on a nonlinear least-squares approach was implemented to correct the I/Q signal of the radar. Then, we have presented a machine learning-based binary classification approach to detect the presence of living occupants in a car using DC-removed radar signals, as a light dataset considering the distinguishable variance pattern of the distribution of alive and non-living classes. The proposed method leverages efficient feature extraction from the radar's complex signal, focusing on the amplitude of the $|I_c[n] + jQ_c[n]|$ signal to distinguish between living and non-living objects. Afterwards, the breathing rate estimation in life class is investigated.

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